



OrBit User Guide

Module 02: Local-First Code Synchronization & P2P Pushing

PUBLIC DOCUMENTATION — FOR DEVELOPERS USERS

1 Overview of Code Synchronization

In traditional software development workflows, synchronization relies on centralized Git host remotes (e.g., GitHub, GitLab) where developers manually stage, commit, and push commits. OrBit introduces a **local-first continuous sync paradigm**:

- Code changes are continuously monitored by the local Rust daemon.
- Synchronization events compute high-performance diffs using the **sync-diff** engine.
- Diffs are encrypted end-to-end (E2EE) and distributed directly over a peer-to-peer (P2P) mesh network using **libp2p**.

2 The Push Workflow

2.1 1. Previewing Diffs (Pre-Push Review)

Before broadcasting code changes to your team, OrBit allows you to review modified, added, and removed files:

1. Open the OrBit Desktop Client or press **Ctrl+Shift+P** to launch the HUD overlay.
2. Click **Preview Push**.
3. The client invokes the Rust daemon command **preview_push**, comparing your current workspace tree against the baseline **main** snapshot.
4. Inspect the file list and diff counts in the **Pre-Push Review Modal**.

2.2 2. Executing a Peer-to-Peer Push (local_push)

When you click **Confirm Push**:

1. The local daemon takes a snapshot of your workspace.
2. It computes a Blake3 delta payload containing changed file contents.
3. The delta is signed with your local Ed25519 identity key and encrypted using ChaCha20Poly1305 with your project’s secret token.
4. The daemon broadcasts the encrypted delta directly to active peers over the libp2p mesh network.

P2P Bypass

Direct peer-to-peer pushes bypass the central Render server entirely, guaranteeing zero cloud storage of your unencrypted source code and eliminating bandwidth bottlenecks.

2.3 3. Encrypted Cloud Relay Fallback (full_push)

If all target peers are currently offline when you push changes:

1. The local daemon detects zero connected online peers.
2. It uploads the E2EE encrypted delta blob to the Go Control Server’s **Dead-Drop Vault** (POST /api/v1/projects/{id}/push).
3. When offline peers reconnect, their client queries GET /api/v1/projects/{id}/pull to download and apply pending deltas.
4. All cloud relay blobs are automatically purged after 7 days by an automated background sweeper process.

3 Sync Status Indicators

The OrBit UI provides real-time visual indicators to keep you informed of synchronization health:

Status Badge	Meaning & Description
Synced	Local sub branch is fully synchronized with team main baseline.
Ahead (Pending Push)	Local unpushed changes exist; click Push to broadcast.
Conflict Detected	Incoming peer delta collides with local edits; action required.
Offline	Daemon cannot connect to P2P network or control server.

Table 1: OrBit Synchronization Status Badges